Levi Neuwirth

RESEARCH ENGINEER — TECHNICAL AI ASSURANCE

ln@levineuwirth.org · +1 (845) 332-0898 · levineuwirth.org · GitHub · LinkedIn

Research engineer working on verifiable inference and frontier-model evaluations. Combines cryptographic verifier hardening and GPU workload profiling with evaluator calibration, agent failure analysis and reproducible experiments.

RESEARCH & ENGINEERING EXPERIENCE

Cambridge AI Safety Hub — *MARS V Fellow*
July 2026 – Present · Oxford, United Kingdom

VerInf: zero-knowledge LLM inference research with James Petrie (Future of Life Institute). Cited as prior work by Anthropic (November 2026).

- Built and upstreamed a Python workload profiler, versioned manifests, execution DAGs and cost models; validated workload accounting against an archived Llama-4 Maverick run and calibrated on NVIDIA B200 hardware.
- Integrated a collaborator's expert-weight bridge; fixed enrollment-identity binding, Merkle-index checks and sumcheck-round enforcement in Python and Rust verification paths.
- Implemented targeted causal softmax, SiLU and surprisal repairs with a 10-case GPU regression suite; hardened the Rust-verifier harness to distinguish explicit verdicts from crashes or missing output.

VerInf · Case study

Independent Research Contractor — *Frontier AI labs*
2025 – Present

Lead frontier-model evaluation-environment design and evaluator calibration.

- Build tasks and scoring rubrics in **Harbor**; execute models and debug agent scaffolds and trajectories.
- Identify rubrics that measure something other than the intended task; refine evaluation design before scaling runs. Distinguish model failures from defects in task design, scaffolding and scoring.
- Analyze reasoning, capability, safety and red-team failures. Client identities and specific results are confidential.

Proof Broker — Independent Research *2026 – Present.* Built an OCaml broker connecting Lean 4 and Rocq to untrusted solvers and model-generated search through versioned evidence schemas, independent certificate checks and kernel-checked proof terms.
Built reproducible Python workflows with pinned tasks, deterministic baselines and negative controls.
Audited context dependencies in proofs; repaired integer-goal reconstruction and the final normalizer. Offline replay produced kernel-checked proofs from 36/36 retained certificates at four obligations with no added axioms.

Shu Laboratory, Brown University Department of Neurology — *Undergraduate Researcher and Technical Lead*
October 2023 – May 2026 · Providence, RI

- Co-led Deep Sets outcome models developed on an 80M+ discharge cohort; readmission AUROC was 0.750 versus 0.655 for the Charlson index on 3.23M temporally held-out discharges. Deployed the public calculator with a Python/FastAPI backend; manuscript under review at JAMIA.
- Technical lead for NeuroPose across four externally funded projects: Python/TensorFlow inference, validated outputs, persistent job state across crashes, Docker deployment and Rust-backed analysis.

EDUCATION

Brown University — *Sc.B., Mathematics and Computer Science*
August 2022 – May 2026 · Providence, RI

GPA 3.8/4.0. Individual research grant: Brown Undergraduate Teaching and Research Award (UTRA), Summer 2024.

TECHNICAL SKILLS

Languages: Python, Rust, C, C++, Go.
Evaluation: Harbor, Evaluator calibration, Agent scaffolds, Red-teaming.
ML and data: PyTorch, TensorFlow, NumPy, Pandas.
Verification: Lean 4, Rocq, OCaml, SMT (cvc4, cvc5, Z3).
Platforms (Linux): Git, Docker, SLURM.